

Week 10

Perception

Jaerock Kwon, PhD

Associate Professor

Electrical and Computer Engineering

University of Michigan-Dearborn

Today

0:00 – 0:50	Lecture
1:00 – 2:00	Track milestone demos — 8 min per team
2:00 – 2:30	Track work
2:30 – 3:00	Standup — Merge 1 next week

1. What the sensors measure

2D LiDAR

Range to the nearest surface, along a set of rays,
in a single horizontal plane. That is all.

Samples	720 (0.5° resolution)
Field of view	360°
Range	12 m
Rate	10 Hz

What a horizontal-plane scanner cannot see

- **Below the plane** — a kerb, a step, a hole
- **Above it** — a table edge, a low shelf, an open drawer
- **Transparent or mirrored** — glass doors read as open space

And on this vehicle: the LiDAR **faces forward**.

There is no rear sensing at all.



The map does not distinguish **nothing there** from **nothing visible**.

SLAM and Nav2 both treat "no return" as free space.

Camera



Colour and texture, projected onto a plane. **No depth.**

Resolution	640 × 480
Horizontal FOV	1.396 rad (~80°)
Rate	30 Hz

Rolling shutter

The sensor exposes the image **row by row**, not all at once.

Move during the scan and the image **shears** —
vertical lines lean, and the lean scales with speed.

The ADR chooses a **\$30 rolling-shutter USB camera** for semester 1.

Why that is a good decision, not a cheap one

At $\leq$ walking pace, on a vehicle whose limiting factor is **feedback quality** rather than image quality, shear is small.

And the ADR names what would flip it:

Phase 2 is behavior cloning. The camera stops being something you look at and becomes the **input to a learned policy**. A speed-dependent geometric distortion in the training data is then a systematic bias the network will happily learn — and it will not show up in the loss curve.

2. Calibration

Intrinsics — the camera's own geometry

Focal length, principal point, lens distortion.

The mapping from **3D rays to pixels**.

Without it you cannot turn a pixel into a direction,
so you cannot relate the image to anything else in the world.

Procedure: a checkerboard, many views, varied distance and angle.

The twin cannot teach you this

In simulation the camera is a **perfect pinhole** and its `camera_info` is exact.

So intrinsic calibration is free there — not because it is easy, but because **it does not exist as a problem.**

Worth noticing rather than enjoying.

Extrinsics — where the sensor **is**

Intrinsics are about the sensor. Extrinsics are about the **vehicle**.

In simulation you typed the numbers into the URDF and they became true.

On hardware it runs the other way: the sensor is bolted somewhere, and you must **measure** where.

The staged procedure

Stage	
Coarse	Ruler from the <code>base_link</code> origin. Good to ~1 cm
LiDAR refinement	Drive parallel to a flat wall; the scan must be parallel too
Camera → LiDAR	Project scan points into the image; they must land on their objects

Lab 5 already showed you the cost

A **5 cm** error. Nothing errors. LiDAR publishes, TF is complete, SLAM runs — and walls observed from different headings stop landing on top of each other.

A **constant** mounting offset makes a **heading-dependent** map error, because that offset points in a different world direction as the car rotates.

Which is why the gate is stated as "**the same wall within 10 cm**" rather than as something about the sensor.

Time sync is not a formality

If the LiDAR's timestamps are offset from the vehicle's, TF happily interpolates a pose for the **wrong instant**.

Every scan is placed slightly wrong — **in a way that scales with speed**.

So it looks like a mechanical problem, not a clock problem.

(This is also why ADR-SW1 insists on one transport and one clock.)

Track milestone demos — 8 min each

Show one thing working end to end, **with a number attached**.

Track	A good answer
Electrical	Steering holds an angle — here is the steady-state error and loop rate
Chassis	Measured wheelbase, track, wheel radius — and how each was measured
Simulation	<code>mitt_bench</code> reports odometry drift over 20 m — here is the number
Software	<code>mitt_hardware</code> loads, activates, round-trips a recorded bag

Merge 1 is next week

Electrical + Chassis: the vehicle steers to a commanded angle, wheels off the ground. Closes bring-up **Stage 1**.

Anything that will not be ready must be raised **today**.

Chassis: wheel radius is measured **loaded, on the floor**.

Hollow plastic wheels deflect under 6 kg of payload.